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HW 6:

Exercise Section 8.1, Problem 4 (vii) and (ix), including (a) to (f). That is, you do not need to work on (g) and (h).

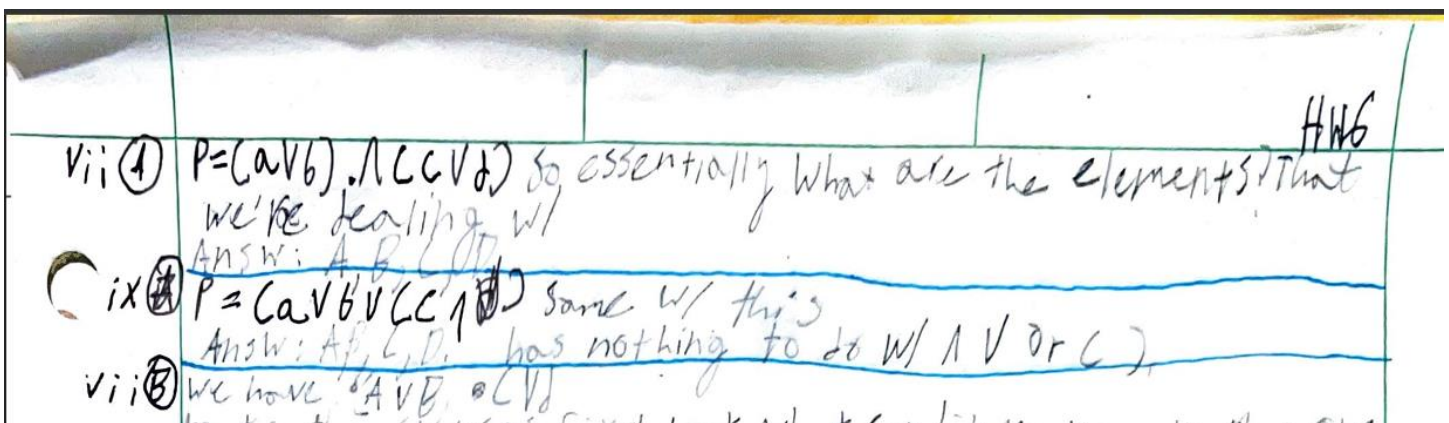
HW6

Use predicates (vii) and (ix) to answer the following questions.

vii. $p = (a \vee b) \wedge (c \vee d)$

ix. $p = a \vee b \vee (c \wedge d)$

(a) List the clauses that go with predicate p .



(b) Compute (and simplify) the conditions under which each clause determines predicate p .

Ans: A, C, D has nothing to do w/ $A \vee$ or C .

vii) We have $A \vee B, C \vee D$
 make the clauses fixed, look what condit make only that one clause control the outcome of P
 $a: I$ want a to be the only clause that can flip P
 $b \rightarrow \text{false}$ $a \vee b$ depends only on a
 $C \vee D \rightarrow$ must be true thus $\neg B \wedge (C \vee D)$
 $\bullet a \wedge (C \vee D) \rightarrow P$
 $\bullet \neg a \wedge (C \vee D) \rightarrow \neg P$ } separate the other group false, & oppo true.
 $\bullet C \wedge a \vee D \rightarrow P$

ix) Same thing practically, so that you have $O \vee E$
 isolate the clause, fix the others that only affects result
 $P: I$ want a making the differ. Thus, b must be false.
 $\& C \vee D = \text{false}$ too b/c C or $D = \text{either}$ is false.
 $\bullet \neg a \wedge (C \vee D)$
 $\bullet \neg a \wedge \neg b \wedge (C \vee D)$
 $\bullet \neg a \wedge \neg b \wedge \neg C \wedge \neg D$, both a & b must be $\neg$ C depends on D
 $\bullet \neg a \wedge \neg b \wedge \neg C$ & the other lit in conjunction must be true

(viii) $4 \times 4 = 16$ rows, for both vii & ix

- (c) Write the complete truth table for the predicate. Label your rows starting from 1. Use the format in the example underneath the definition of Combinatorial Coverage in Section 8.1.1. That is, row 1 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the value of the predicate itself.

• 7 a 1 7 b 1 d } C & D, both a & b must be 1 + C depends on D
 • 7 a 1 7 b 1 c } & the other lit in conjunction must be true

①. $4 \times 4 = 16$ rows for both v_i & i_x

Dealing w: ① $v_i (A \vee B) \wedge (C \vee D)$ ② $i_x \in A \vee B \vee (C \wedge D)$

	A	B	C	D	P	P _a	P _b	P _c	P _d
1	T	T	T	T	F	F	F	F	F
2	T	T	T	F	F	F	F	F	F
3	T	T	T	F	F	F	F	F	F
4	T	T	F	F	F	F	F	F	F
5	T	F	T	T	F	F	F	F	F
6	T	F	T	F	F	F	F	F	F
7	T	F	T	F	F	F	F	F	F
8	T	F	F	F	F	F	F	F	F
9	F	T	T	T	F	F	F	F	F
10	F	T	T	F	F	F	F	F	F
11	F	T	F	T	F	F	F	F	F
12	F	T	F	F	F	F	F	F	F
13	F	F	T	T	F	F	F	F	F
14	F	F	T	F	F	F	F	F	F
15	F	F	F	T	F	F	F	F	F
16	F	F	F	F	F	F	F	F	F

④ $A \vee B$
 ⑤ $C \vee D$
 Then do
 & in between
 to get the
 pos P_b, P_c,
 P_d col

③ a
 b
 $C \wedge D$
 Then for P_a,
 P_b, P_c, P_d I_a,
 use the
 expression 3. From
 part 6 &
 check out
 each row
 to see if
 condit matches
 Yes = T No = F

	A	B	C	D	P	P _a	P _b	P _c	P _d
1	T	T	T	T	F	F	F	F	F
2	T	T	T	F	F	F	F	F	F
3	T	T	F	T	F	F	F	F	F
4	T	T	F	F	F	F	F	F	F
5	T	F	T	T	F	F	F	F	F
6	T	F	T	F	F	F	F	F	F
7	T	F	T	F	F	F	F	F	F
8	T	F	F	F	F	F	F	F	F
9	F	T	T	T	F	F	F	F	F
10	F	T	T	F	F	F	F	F	F
11	F	T	F	T	F	F	F	F	F
12	F	T	F	F	F	F	F	F	F
13	F	F	T	T	F	F	F	F	F
14	F	F	T	F	F	F	F	F	F
15	F	F	F	T	F	F	F	F	F
16	F	F	F	F	F	F	F	F	F

- (d) List all pairs of rows from your table that satisfy General Active Clause Coverage (GACC) with respect to each clause.

(vii) $P = (A \vee B) \wedge (C \vee D)$

Look at previous answer for both vii & ix.
use truth table & scanned for row pairs where
a flip

b, c, d = the same

p flips too

For a the GACC pairs rows [5, 6, 7] x [3, 14, 15]

These are all combos where 1 var has a diff & the other
a = F, & that switch makes p flip & clause, hold others,
repeat b, c, d for that logic flip 1 clause, hold others,
verify p flips, then to how you get GACC.

(ix) $P = (A \vee B) \vee (C \wedge D)$

same thing Find rows where

Just the clause change

everything stays fixed

The clause affects the value of p

For a: GACC pairs are [6, 7, 8] [14, 15, 16]

All these pairs differ only in a & flip the val of p

Checking for cause & effect by flipping 1 clause &

seeing if it's enough to change the predicate

Answer for both: GACC pairs for clauses a, b, c, & d are

- (e) List all pairs of rows from your table that satisfy Correlated Active Clause Coverage (CACC) with respect to each clause.

(viiE)

Answer for both: CACC pairs for clauses a, b, c, & d are
the same as GACC pairs.

(ixE)

it's b/c we're dealing w/ T or F which is practically
binary so flipping them would be the same. Also
the main problem is we are dealing w/ a, b, c, d. Nothing
of p. p flips on its own when we mess w/ them
at least just replace a w/ p

(viiF)

so RACC: C from the book

- (f) List all pairs of rows from your table that satisfy Restricted Active Clause Coverage (RACC) with respect to each clause.

ViiF

so RACC: C from the book
1) Strict Clause Coverage 1) Clause Flips between 2 rows
2) only that clause flips - all other clauses (minor ones) = identical
3) Predicate also flips
Look at table in Pt C.

Let's do this for V, i
Rows (5, 13), (6, 14), (7, 15)
in each of these, only a change, & that causes
P to change too.
a: (5, 13), (6, 14), (7, 15) b: (3, 4), (7, 8), (11, 12)
c: (9, 13), (10, 14), (11, 15)
d: (2, 4), (6, 8), (10, 12)

for each clause I just found all matching row pairs
where exactly 1 bit flips, & it flips as well.

ixF

some RACC Look at table in Pt C

so for
a: (5, 14), (7, 15), (8, 16)
b: (10, 14), (11, 15), (12, 16)
c: (2, 4), (6, 8), (10, 12)

c: C & D it's the same as CACC, only 1 flip